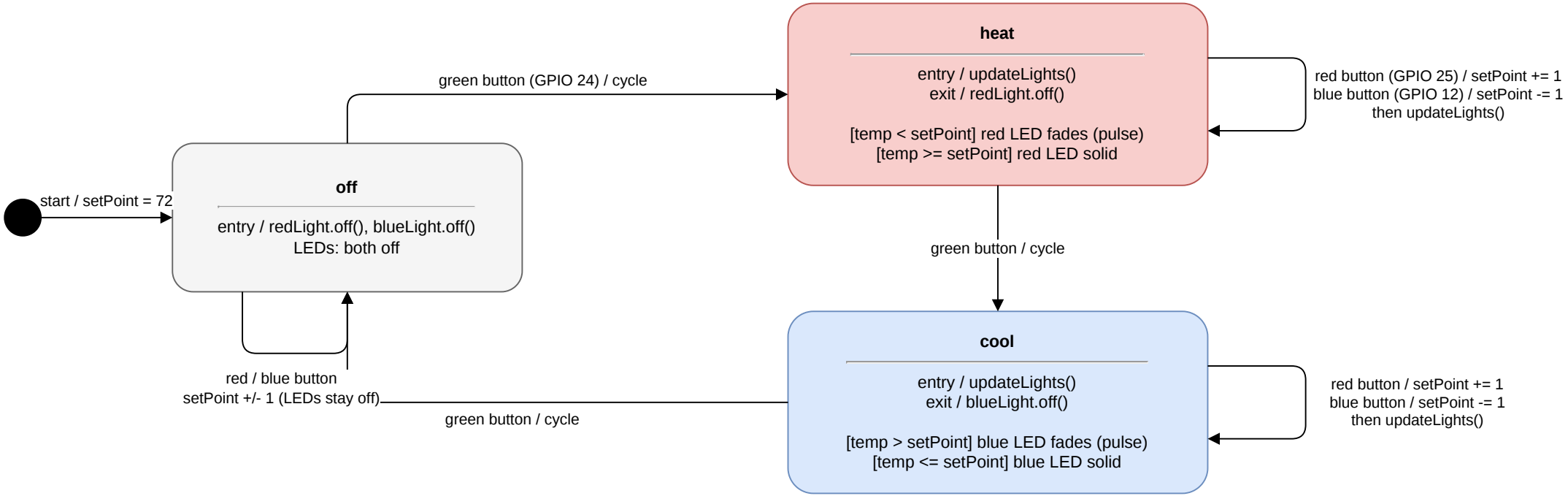
TemperatureMachine State Machine (Smart Thermostat Prototype)

CS 350 Final Project | Luca Formento



Concurrent activities and peripherals

- **Display thread (every 1 s):** LCD line 1 = date and time; line 2 alternates every 5 s between "Temp: 71.6F" and "HEAT Set: 72F". updateLights() re-runs every 10 s so the LEDs follow the room temperature.
- **UART (every 30 s):** sends "state,temperature F,setPoint" (e.g. heat,71.60,72) on /dev/ttyS0 at 115200 baud, 8N1.
- **I2C:** AHT20 sensor read with adafruit_ahtx0 (address 0x38); Celsius converted to Fahrenheit, floor() used for comparisons.
- **GPIO:** red LED GPIO 18, blue LED GPIO 23 (PWMLED); buttons on GPIO 24, 25, 12 use gpiozero when_pressed interrupts.
- **Ctrl-C:** ends the display thread, clears the LCD, and gpiozero releases the GPIO lines.